```
medical_text = "The patient presented with acute myeloid leukemia. Treatment involved chemotherapy
print("Medical text corpus loaded successfully.")
```

```
Medical text corpus loaded successfully.
```

```
import nltk
import spacy

try:
    nltk.data.find('tokenizers/punkt')
except LookupError:
    nltk.download('punkt')
    print("NLTK 'punkt' tokenizer data downloaded.")
else:
    print("NLTK 'punkt' tokenizer data already downloaded.")

try:
    nlp = spacy.load('en_core_web_sm')
except OSError:
    print("Downloading spaCy model 'en_core_web_sm'...")
    !python -m spacy download en_core_web_sm
    nlp = spacy.load('en_core_web_sm')
    print("spaCy model 'en_core_web_sm' downloaded and loaded.")
else:
    print("spaCy model 'en_core_web_sm' already loaded.")

print("NLTK and spaCy libraries initialized.")
```

```
[nltk_data] Downloading package punkt to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt.zip.
NLTK 'punkt' tokenizer data downloaded.
spaCy model 'en_core_web_sm' already loaded.
NLTK and spaCy libraries initialized.
```

```
import nltk
import spacy

try:
    nltk.data.find('tokenizers/punkt')
except LookupError:
    nltk.download('punkt')
    print("NLTK 'punkt' tokenizer data downloaded.")
else:
    print("NLTK 'punkt' tokenizer data already downloaded.")

# Attempt to download 'punkt_tab' as suggested by the error message from the previous execution
try:
    # Check if 'punkt_tab' can be found before attempting to download
    nltk.data.find('tokenizers/punkt_tab')
except LookupError:
    print("NLTK 'punkt_tab' tokenizer data not found, attempting download as suggested...")
    try:
        nltk.download('punkt_tab')
        print("NLTK 'punkt_tab' tokenizer data downloaded.")
    except Exception as e:
        print(f"Failed to download 'punkt_tab': {e}. This resource might not be directly availab
else:
    print("NLTK 'punkt_tab' tokenizer data already downloaded.")
```

```
try:
    nlp = spacy.load('en_core_web_sm')
except OSError:
    print("Downloading spaCy model 'en_core_web_sm'...")
    !python -m spacy download en_core_web_sm
    nlp = spacy.load('en_core_web_sm')
    print("spaCy model 'en_core_web_sm' downloaded and loaded.")
else:
    print("spaCy model 'en_core_web_sm' already loaded.")

print("NLTK and spaCy libraries initialized.")
```

```
NLTK 'punkt' tokenizer data already downloaded.
NLTK 'punkt_tab' tokenizer data not found, attempting download as suggested...
[nltk_data] Downloading package punkt_tab to /root/nltk_data...
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.
NLTK 'punkt_tab' tokenizer data downloaded.
spaCy model 'en_core_web_sm' already loaded.
NLTK and spaCy libraries initialized.
```

```
from nltk.stem import PorterStemmer, LancasterStemmer

# Initialize stemmers
porter_stemmer = PorterStemmer()
lancaster_stemmer = LancasterStemmer()

# Initialize lists to store stemmed words
porter_stemmed_words = []
lancaster_stemmed_words = []

# Process words from nltk_word_tokens
for sentence_tokens in nltk_word_tokens:
    for word in sentence_tokens:
        # Convert word to lowercase before stemming for better consistency
        word_lower = word.lower()
        porter_stemmed_words.append(porter_stemmer.stem(word_lower))
        lancaster_stemmed_words.append(lancaster_stemmer.stem(word_lower))

# Prepare words for display examples
example_original_words = ['presented', 'involved', 'complicated', 'leukemia', 'transplantation',

print("--- Porter Stemmer Examples ---")
print("Original        Porter Stemmed")
print("------------------------------")
for word in example_original_words:
    print(f"{word:<15} {porter_stemmer.stem(word.lower())}")

print("--- Lancaster Stemmer Examples ---")
print("Original        Lancaster Stemmed")
print("-----------------------------------")
for word in example_original_words:
    print(f"{word:<15} {lancaster_stemmer.stem(word.lower())}")

print("Stemming applied and examples displayed.")
```

```
--- Porter Stemmer Examples ---
Original        Porter Stemmed
------------------------------
presented       present
involved        involv
```

```
complicated     complic
leukemia        leukemia
transplantation transplant
recovery        recoveri
patient         patient
acute           acut
myeloid         myeloid
chemotherapy    chemotherapi
severe          sever
neutropenia     neutropenia
--- Lancaster Stemmer Examples ---
Original        Lancaster Stemmed
-----------------------------------
presented       pres
involved        involv
complicated     comply
leukemia        leukem
transplantation transpl
recovery        recovery
patient         paty
acute           acut
myeloid         myeloid
chemotherapy    chemotherapy
severe          sev
neutropenia     neutropen
Stemming applied and examples displayed.
```

```
import spacy

# Ensure the nlp object is loaded from previous steps
# If it wasn't loaded, we'd need to load it here again
try:
    nlp = spacy.load('en_core_web_sm')
except OSError:
    print("spaCy model 'en_core_web_sm' not found. Please run the installation step first.")
    # Exit or raise error if model is not available

# Process the medical text with spaCy (assuming 'medical_text' is defined)
# If 'doc' from previous steps is not in scope, re-process the text.
if 'doc' not in locals() or 'doc' not in globals():
    doc = nlp(medical_text)

# Apply lemmatization
spacy_lemmas = [token.lemma_ for token in doc]

# Prepare words for display examples from the processed doc for direct comparison
# Filter out punctuation and short words that don't clearly demonstrate lemmatization
example_lemmatization_words = []
for token in doc:
    if token.is_alpha and len(token.text) > 2 and token.text.lower() not in [item.lower() for item
        example_lemmatization_words.append(token.text)
    if len(example_lemmatization_words) >= 10:
        break

print("--- spaCy Lemmatization Examples ---")
print("Original        Lemmatized")
print("--------------------------------")
for word in example_lemmatization_words:
    # Find the corresponding token in the doc and get its lemma
    # This approach ensures we are using the lemma derived from spaCy's processing of the original
    found_lemma = ""
    for token in doc:
        if token.text.lower() == word.lower():
```

```
                found_lemma = token.lemma_
                break
        print(f"{word:<15} {found_lemma}")

    print("Lemmatization applied and examples displayed.")
```

```
--- spaCy Lemmatization Examples ---
Original        Lemmatized
---------------------------------
The             the
patient         patient
presented       present
with            with
acute           acute
myeloid         myeloid
leukemia        leukemia
Treatment       treatment
involved        involve
chemotherapy    chemotherapy
Lemmatization applied and examples displayed.
```